

Michael Mihaley

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EDUCATION

Columbia University

Bachelor of Science in Computer Science

New York, NY

Sep. 2024 - May 2027

City College of New York (Macaulay Honors College)

Bachelor of Science in Computer Science, Dean's List

New York, NY

Sep. 2023 - May 2024 (transferred)

EXPERIENCE

Robotics and Reinforcement Learning Researcher

May 2026 – Ongoing

New York Institute of Technology, Advisors: [Feng Han](#)

New York, NY

- Developing a PPO-based reinforcement learning policy for grasping and manipulating deformable objects of unknown stiffness using only gripper force feedback
- Integrating differentiable PID layers and slip feedback via bilateral force readings into policy network, targeting object slip rate per force applied reduction of 40% relative to a finite-state-machine baseline
- Building a depth-based grasp targeting pipeline with *Intel RealSense D435* localization to estimate object pose and determine grasp locations

Robotics and Imitation Learning Researcher

May 2025 – Sep. 2025

Columbia University, Advisors: [Yunzhu Li](#) and [Yixuan Wang](#)

New York, NY

- Implemented Python/C++ control pipelines for 2 robotic systems (*Dexmate Vega*, *xArm7*), supporting joint-space and IK-based end-effector control with sub-cm positioning error
- Designed leader arm and extended the *GELLO Teleoperation Framework* to perform imitation learning with the *Dexmate Vega*, simulating in *SAPIEN* and supporting 500 Hz control loops
- Designed and printed custom mounting connections to expand end effector options for the *Dexmate Vega* by 50%

Adjunct Faculty, Robotics

Oct. 2025 – Present

Rudolf Steiner School

New York, NY

- Instructing high school students on advanced programming concepts such as object-oriented design and sorting algorithms
- Mentoring a *First Tech Challenge* robotics team across mechanical design, electronics, and Java programming, guiding them to achieve competition-ready results
- Scaled program size from 2-3 students to around 15 in one semester through hands-on curriculum design

Machine Learning Intern

January 2024 – June 2024

Outlier AI

Remote

- Refined large language models via reference-free evaluation in Python for a response accuracy increase of over 35%
- Built automated safety checks combining classifiers and rule tests to flag 10% of pre-deployed output as misaligned

PROJECTS

Performance Optimized LLM Inference for Real-Time Robotic Planning | *Python, CUDA*

- Built a robotics planning pipeline using *Qwen2.5-3B-Instruct* to convert natural language commands into action sequences, conducted full-stack GPU profiling (compute vs memory-bound analysis).
- Utilized 16-bit quantization on weights and activations, VLLM prefix caching to reduced end-to-end latency by 60%, increased tokens/sec by 3×, and decreased VRAM usage by > 30%. Benchmarked run telemetry via *Weights and Biases*.
- Integrated LLM inference with *ROS2* and *Gazebo* simulation for performing generated action sequences.

HybVIO Visual-Inertial Odometry System with SLAM | *C++, C*

- Extended a research-oriented odometry system that uses preprocessed data to interface with actual *Intel RealSense* Cameras for real-time spatial mapping under constrained compute (Quad-core, 1.2GHz clock speed)
- Collaborated with a team to identify slowdowns in stereo imaging code and speed up visualization by 60%

TECHNICAL SKILLS

Languages: C++, Java, CUDA, Python, Kotlin, C#

Development Tools: Docker, Android Studio, Fusion 360, SAPIEN, Drake, CMake, MuJoCo, Pytorch, Linux, ROS

Certifications and Fellowships: CAIAC AI Policy Fellow, Coursera Deep Learning Certified